TP Modul 05

Nama : William Peter Vanx Najoan
NIM : 103032400084
Kelas : IT-48-01
Kode Asprak : APY

Soal 1

list.h

```
#ifndef LIST_H_INCLUDED
#define LIST_H_INCLUDED

#include <iostream>

using namespace std;

typedef int infotype;
typedef struct elmlist *address;

struct elmlist {
    infotype info;
    address next;
};

struct List {
    address first;
};

void searchElement_103032400084(List &L, infotype x);
void selectionSort_103032400084(List &L);
void insertSorted_103032400084(List &L, address p);

#endif // LIST_H_INCLUDED

```

list.cpp

```
#include <iostream>
#include "list.h"

using namespace std;

void searchElement_103032400084(List &L, infotype x) {
    address current;
    int position;

    current = L.first;
    position = 0;

    while (current != nullptr && current -> info != x) {
        position++;

        current = current -> next;
    }

    if (current != nullptr) {
        cout << current << " " << position;
    } else {
        cout << "elemen tersebut tidak ada dalam list";
    }
}

void selectionSort_103032400084(List &L) {
    address p, min, temp;
    infotype x;

    p = L.first;
```

```
    while (p != nullptr) {
        min = p;
        temp = p;

        while (temp != nullptr) {
            if (temp -> info < min -> info) {
                min = temp;
            }
            temp = temp -> next;
        }
        x = p -> info;
        p -> info = min -> info;
        min -> info = x;

        p = p -> next;
    }
}

void insertSorted_103032400084(List &L, address p) {
    address q, prev;
    bool found;

    q = L.first;
    found = false;
    prev = nullptr;

    while (q != nullptr && found == false) {
        if (q -> info < p -> info) {
            prev = q;
```

```
            q = q -> next;
        } else {
            found = true;
        }
    }

    if (prev == nullptr) {
        p -> next = q;
        L.first = p;
    } else {
        p -> next = q;
        prev -> next = p;
    }
}
```